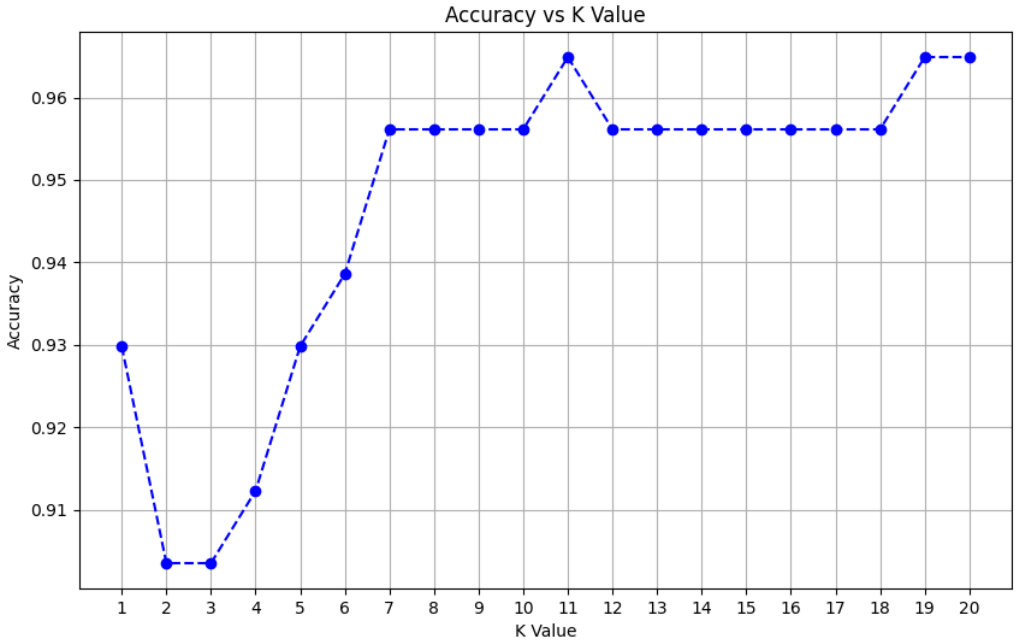
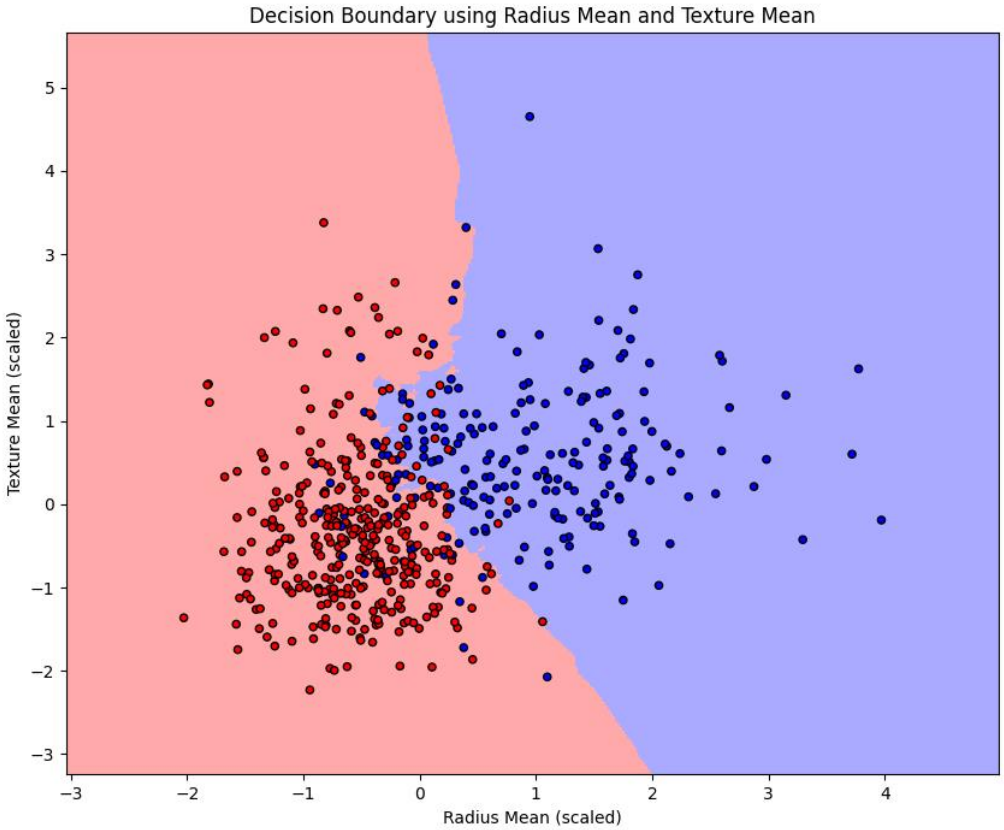
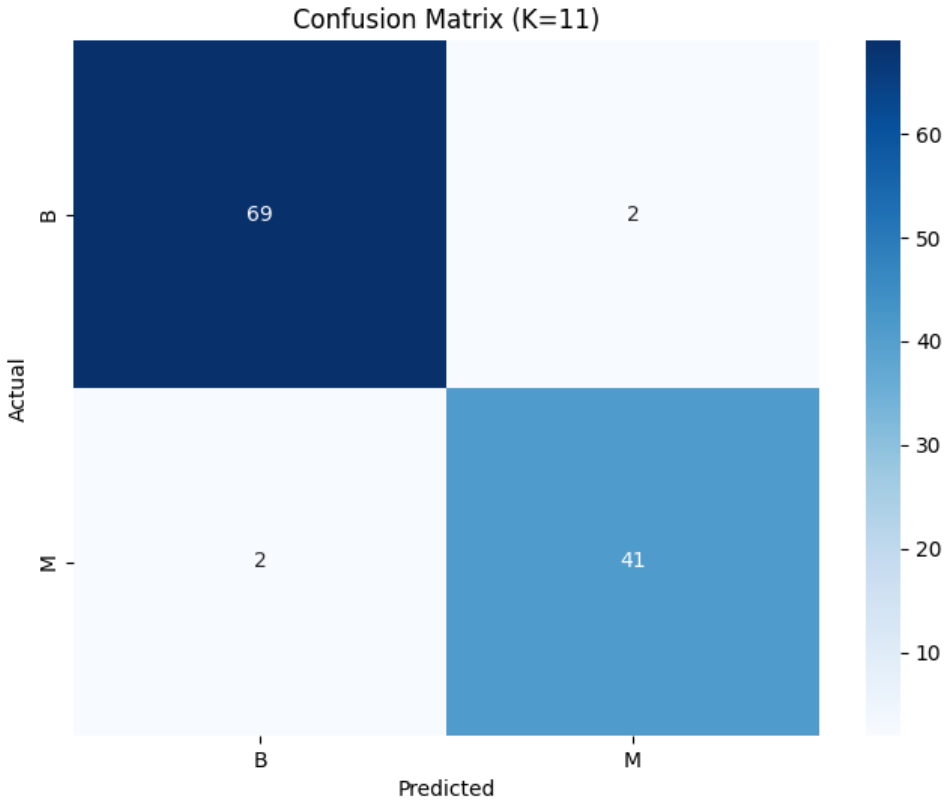
EXPERIMENT-5

SCENARIO-1

```
PS C:\Users\kamal\Downloads> & C:/Users/kamal/AppData/Local/Programs/Python/Python312/python.exe c:/Us
ers/kamal/Downloads/scenario_1_knn.py
--- SCENARIO 1: K-NEAREST NEIGHBORS (KNN) ---

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 32 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   id                       569 non-null    int64
 1   diagnosis                569 non-null    object
 2   radius_mean              569 non-null    float64
 3   texture_mean             569 non-null    float64
 4   perimeter_mean           569 non-null    float64
 5   area_mean                569 non-null    float64
 6   smoothness_mean          569 non-null    float64
 7   compactness_mean         569 non-null    float64
 8   concavity_mean           569 non-null    float64
 9   concave points_mean      569 non-null    float64
 10  symmetry_mean            569 non-null    float64
 11  fractal_dimension_mean   569 non-null    float64
 12  radius_se                569 non-null    float64
 13  texture_se               569 non-null    float64
 14  perimeter_se             569 non-null    float64
 15  area_se                  569 non-null    float64
 16  smoothness_se            569 non-null    float64
 17  compactness_se           569 non-null    float64
 18  concavity_se             569 non-null    float64
 19  concave points_se        569 non-null    float64
 20  symmetry_se              569 non-null    float64
 21  fractal_dimension_se     569 non-null    float64
 22  radius_worst             569 non-null    float64
 23  texture_worst            569 non-null    float64
 24  perimeter_worst          569 non-null    float64
 25  area_worst               569 non-null    float64
 26  smoothness_worst         569 non-null    float64
 27  compactness_worst        569 non-null    float64
 28  concavity_worst          569 non-null    float64
```

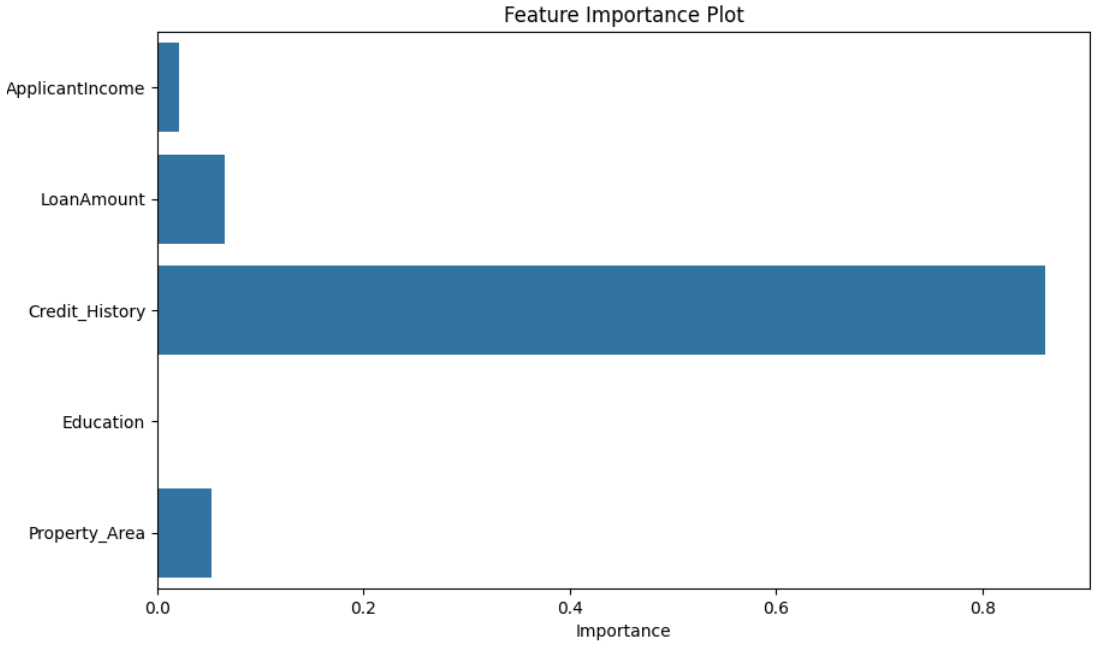

Accuracy vs K Value
Accuracy
K Value
0.96
0.95
0.94
0.93
0.92
0.91
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

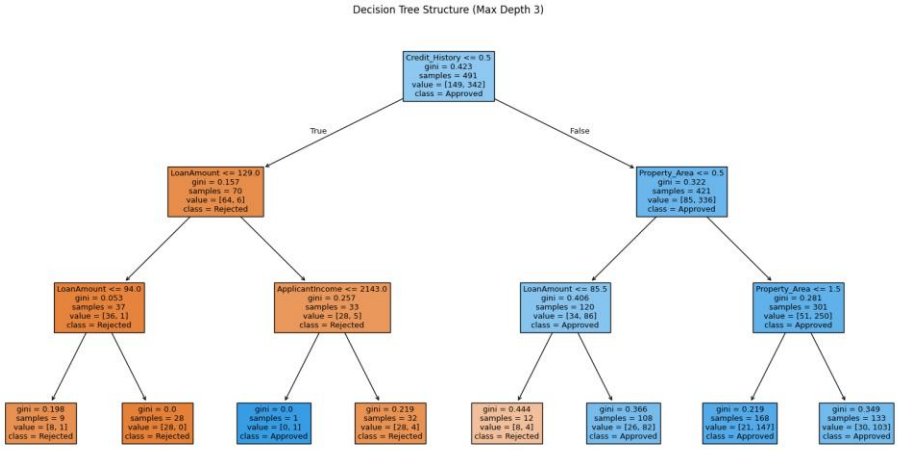

Decision Boundary using Radius Mean and Texture Mean
Texture Mean (scaled)
Radius Mean (scaled)
5
4
3
2
1
0
−1
−2
−3
−3 −2 −1 0 1 2 3 4

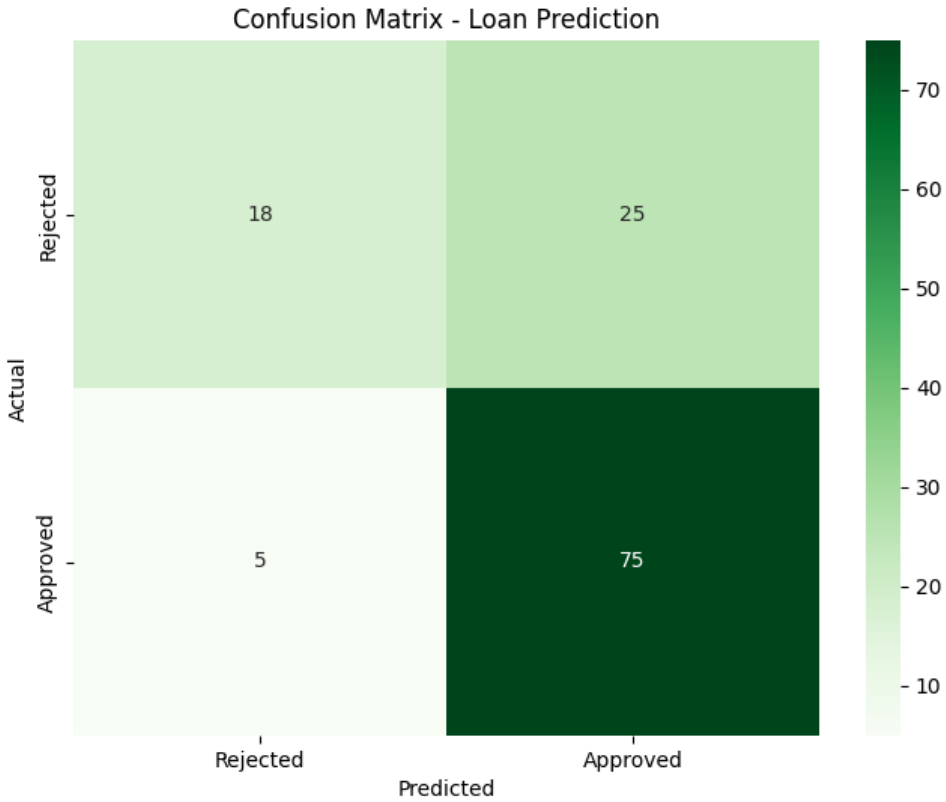

Confusion Matrix (K=11)
69
2
2
41
B
M
Actual
B
M
Predicted
60
50
40
30
20
10

```
 27  compactness_worst        569 non-null    float64
 28  concavity_worst          569 non-null    float64
 29  concave points_worst     569 non-null    float64
 30  symmetry_worst           569 non-null    float64
 31  fractal_dimension_worst  569 non-null    float64
dtypes: float64(30), int64(1), object(1)
memory usage: 142.4+ KB
None

First 5 rows:
         id  ... fractal_dimension_worst
0    842302  ...                 0.11890
1    842517  ...                 0.08902
2  84300903  ...                 0.08758
3  84348301  ...                 0.17300
4  84358402  ...                 0.07678

[5 rows x 32 columns]

Best K identified: 11

Final Model Performance (Best K):
Accuracy: 0.9649
Precision: 0.9535
Recall: 0.9535
F1 Score: 0.9535

Number of misclassified cases: 4
Indices of misclassified cases: [20 39 52 77]
```

SCENARIO-2


Feature Importance Plot
ApplicantIncome
LoanAmount
Credit_History
Education
Property_Area
0.0
0.2
0.4
0.6
0.8
Importance


Decision Tree Structure (Max Depth 3)
Credit_History <= 0.5
gini = 0.423
samples = 491
value = [149, 342]
class = Approved
True
False
LoanAmount <= 129.0
gini = 0.157
samples = 70
value = [64, 6]
class = Rejected
Property_Area <= 0.5
gini = 0.322
samples = 421
value = [85, 336]
class = Approved
LoanAmount <= 94.0
gini = 0.053
samples = 37
value = [36, 1]
class = Rejected
ApplicantIncome <= 2143.0
gini = 0.257
samples = 33
value = [28, 5]
class = Rejected
LoanAmount <= 85.5
gini = 0.406
samples = 120
value = [34, 86]
class = Approved
Property_Area <= 1.5
gini = 0.281
samples = 301
value = [51, 250]
class = Approved
gini = 0.198
samples = 9
value = [8, 1]
class = Rejected
gini = 0.0
samples = 28
value = [28, 0]
class = Rejected
gini = 0.0
samples = 1
value = [0, 1]
class = Approved
gini = 0.219
samples = 32
value = [28, 4]
class = Rejected
gini = 0.444
samples = 12
value = [8, 4]
class = Rejected
gini = 0.366
samples = 108
value = [26, 82]
class = Approved
gini = 0.219
samples = 168
value = [21, 147]
class = Approved
gini = 0.349
samples = 133
value = [30, 103]
class = Approved


Confusion Matrix - Loan Prediction
Actual
Rejected
Approved
18
25
5
75
Rejected
Approved
Predicted
70
60
50
40
30
20
10